ECOLOGY

71. Which one of the following is the correct sequence of a food chain ?
(a) Diatoms-Crustaceans-Herrings
(b) Crustaceans –Diatoms-Herrings
(c) Diatoms-Herrings-Crustaceans
(d) Crustaceans –Herrings-Diatoms

Ans. (a): Diatoms (autotrophic) → Crustaceans (herbivorous) → Herrings (carnivorous). This is the correct sequence of the food chain. Omnivores in the food chain receive high nutrition.

72. There is some concern regarding the nanoparticles of some chemical elements that are used by the industry in the manufacture of various products. Why?
1. They can accumulate in the environment and contaminate water and soil.
2. They can enter the food chains.
3. They can trigger the production of free radicals.
Select the correct answer using he code given below :
(a) 1 and 2 only (b) 3 only
(c) 1 and 3 (d) 1, 2 and 3

Ans. (d): Many of the nano particles are very stable and hence have a tendency for accumulation. They can easily enter food chains because of their very small size and stay in them because of their stable nature. They have the ability to generate free radicals and cause oxidative stress in cells.

73. In India, the problem of soil erosion is associated with which of the following?
1. Terrace cultivation 2. Deforestation
3. Tropical climate
Select the correct answer using the codes given below :
(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): Soil erosion is the result of human activities against nature. The roots of plants keep the soil tight, due to which soil weathering is not possible easily, but where deforestation has taken place, soil erosion is occurring as a result. This problem has taken a serious form in India.

74. Consider the following international agreements
1. The International Treaty on Plant Genetic Resource for Food and Agriculture.
2. The United Nations Convention to Combat Desertification.
3. The World Heritage Convention.
Which of the above has/have a bearing on the biodiversity ?
(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): As per United Nations Convention to Combat Desertification (UNCCD)- "dynamic of land, climate and biodiversity are intimately connected, the UNCCD collaborates closely with the other two Rio conventions: the Convention on Biological Diversity (CBD) and the United Nations Framework Convention on Climate Change (UNFCCC), to meet these complex challenges with an integrated approach and the best possible use of natural resources.
The rest of the two options are also connected with biodiversity.

75. If a wetland of international importance is brought under the 'Montreux Record', what does it imply?
(a) Changes in ecological character have occurred are occurring or are likely to occur in the wetland as a result of human interference
(b) The country in which the wetland is located should enact a law to prohibit any human activity within five kilometers from the edge of the wetland
(c) The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity and therefore the cultural diversity therein should not be destroyed.
(d) It is given the status of 'World Heritage Site'.

Ans. (a): The Montreux Record is a register of wetland sites on the list of wetlands of International Importance where changes in ecological character have occurred, or are occurring, or likely to occur as a result of technological developments, pollution as other human interference. It is maintained as part of the Ramsar List. Currently 48 sites where listed in the Montreux Record. Indian sites in the like Keoladeo National Park-Rajasthan, Loktak Lake - Manipur are included in Montreux Record.

76. Consider the following pairs :

Programme/Project		Ministry
1. Drought –prone Area Programme	:	Ministry of Agriculture
2. Desert Development/ Programme	:	Ministry of Environment and Forests
3. National Watershed / Development Project	:	Ministry of Rural Development

Which of the above pairs is/are correctly matched?
(a) 1 and 2 only (b) 3 only
(c) 1,2 and 3 (d) None

Ans. (d): Drought prone Area Programme and Desert Development Programme are being run by the Department of Land Resources, Ministry of Rural Development. The National Watershed Development project for rainfed areas is being run by the Department of Agriculture with the cooperation of its various departments.

77. With reference to Bombay Natural History Society (BNHS), consider the following statement-
1. It is an autonomous organization under the Ministry of Environment and Forests.
2. It strives to conserve nature through action – based research, education and public awareness.
3. It organizes and conducts nature trails and camps for the general public.
Which of the statements given above is/are correct?

(a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1,2 and 3

Ans. (c): The Bombay Natural History Society is one of the largest non-governmental organizations in India, established on 15 September, 1883 engaged in conservation and biodiversity research. In strives to conserve nature through action-based research education and public awareness.

78. With reference to 'Global Environment Facility', which of the following statements is/ are correct ?
(a) It serves as financial mechanism for 'Convention on Biological Diversity' and 'United Nations Framework Convention on Climate Change'.
(b) It undertakes scientific research on environmental issues at global level.
(c) It is an agency under OECD to facilitate the transfer of technology and funds to underdeveloped countries with specific aim to protect their environment.
(d) Both (a) and (b)

Ans. (a): GEF is an independently operating financial organization, the GEF provides grants for projects related to biodiversity climate change, international waters land degradation, the Ozone layer and persistent organic pollutants.

79. Consider the following pairs:
1. Dampa Tiger Reserve : Mizoram
2. Gumti Wildlife Sanctuary : Sikkim
3. Saramati Peak : Nagaland
Which of the above pairs is/are correct matched?
(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Dampa Tiger Reserve is located in Mizoram. The Gumti Wildlife Sanctuary is in Tripura, while the Saramati peak extends from the state of Nagaland in India to the Sagaing region of Burma.

80. With reference to conservation organization called' Wetlands International', Which of the following statements is/ are correct?
1. It is an inter-governmental organization formed by the countries which are signatories of Ramsar Convention.
2. It works at the field level to develop and mobilize knowledge, and use the practical experience to advocate for better policies.
Select the correct answer using the code given below;
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b): Wetlands International is a non-profit and non-governmental organisation working to protect and enhance wetlands and its resources. Its headquarter is located in the Netherlands and the organization operates in over 100 countries.

81. Brominated flame retardants are used in many household products like mattresses and upholstery. Why is there some concern about their use?
1. They are highly resistant to degradation in the environment.
2. They are able to accumulate in humans and animals.
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c): Brominated flame retardants (BFRs) have routinely been added to customer products to make it more difficult for a material to ignite or reduce the spread of fire. But due to their chemical structure, they are highly resistant to degradation in the environment, due to which they can accumulate in humans and animals if entered into the digestive system.

82. Other than poaching, what are the possible reasons for the decline in the population of Ganges River Dolphins?
1. Construction of dams and barrages on rivers.
2. Increase in the population of crocodiles in rivers.
3. Getting trapped in fishing nets accidentally.
4. Use of synthetic fertilizers and other agricultural chemicals in crop-fields in the vicinity of rivers.
Select the correct answer using the code given below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1, 3 and 4 (d) 1, 2, 3 and 4

Ans. (c): The Ganga River Dolphin is found in the Ganges-Brahmaputra-Meghna (GBM) and Karnaphuli-Sangu rivers of India and Bangladesh. The population of the Ganges Rivers Dolphin has declined to less than 1800 from 6000 Since 1982 due to construction of dams and water pollution caused by pesticides, fertilizers and industrial effluents.
The crocodile population has nothing to do with the decline in population of dolphins.

83. Which of the following add/adds carbon dioxide to the carbon cycle on the planet Earth ?
1. Volcanic action
2. Respiration
3. Photosynthesis
4. Decay of organic matter
Select the correct answer using the code given below
(a) 1 and 3 only (b) 2 only
(c) 1, 2 and 4 (d) 1, 2,3 and 4

Ans. (c): Activities such as volcanic activity, respiration and decay of organic matter etc, contribute carbon dioxide to the carbon cycle on the Earth's surface. In the process of photosynthesis, oxygen is released by plants by consuming carbon dioxide.

84. If you walk through countryside, you are likely to see some birds stalking alongside the cattle to seize the insects disturbed by their movement through grasses. Which of the following is/are such bird/ birds?
1. Painted Stork
2. Common Myna
3. Black –necked Crane
Select the correct answer using the code given below :